

NAME SCHOOL TEACHER 

Pre-Junior Certificate Examination, 2018

Mathematics

Paper 2 Ordinary Level

Time: 2 hours

300 marks

School Stamp

Running Total	
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For Examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3			
4			
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 12 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if you do not show all necessary work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1 (Suggested maximum time: 5 minutes)

Question 1 (Suggested maximum time: 5 minutes)

Claire is attending a two-day sports camp next weekend.

She can take part in one sport on each day.

The different sports she can select each day are shown in the table.

Karl says “Claire has 9 possible options when selecting her sports”.

Saturday	Sunday
Golf	Camogie
Soccer	Tennis
Basketball	Swimming

(a) Show that he is correct.

[illegible]

(b) Complete the table to show the 9 possible options.

	Saturday	Sunday
1	Golf	Camogie
2		
3		
4		
5		
6		
7		
8		
9		

Claire selects at random the sports she will take part in.

(c) Find the **probability** that Claire selects soccer and swimming.

[illegible]

Answer =

(d) Find the **probability** that Claire selects golf as one of her options.

[illegible]

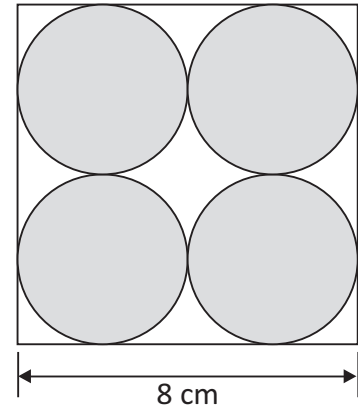
Answer =

Question 2

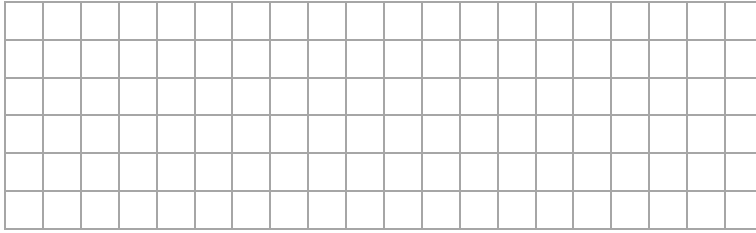
(Suggested maximum time: 10 minutes)

The design of a logo has four grey identical discs within a white square as shown.

The square has a side of length 8 cm.



- (a) Find the **area** of the square.

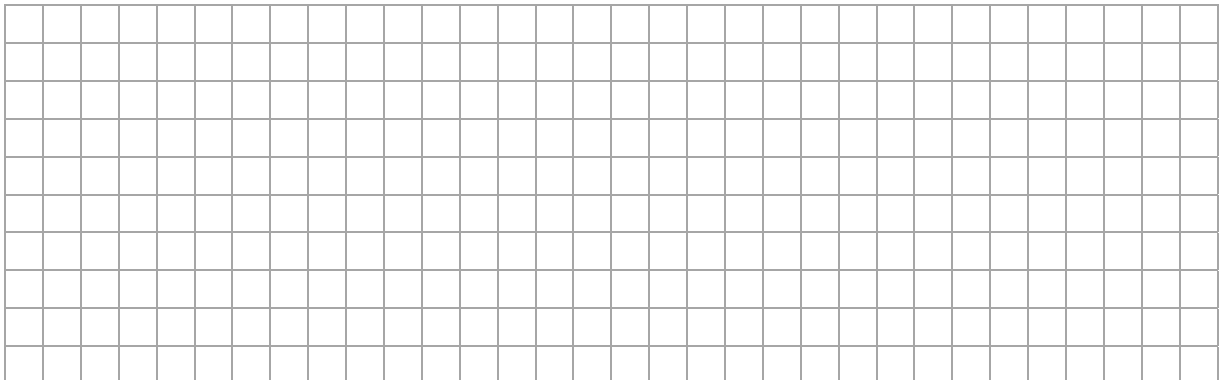


- (b) Kim says “the length of the radius of each disc is 2 cm”.
Is she correct? Justify your answer fully.

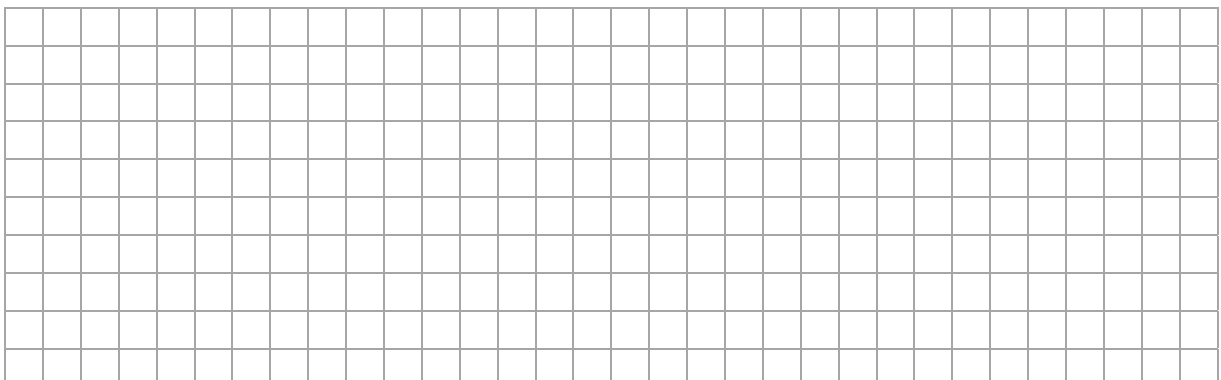
Answer: _____

Justification: _____

- (c) Find the **area** of one of the discs.
Give your answer in cm^2 , correct to one decimal place.

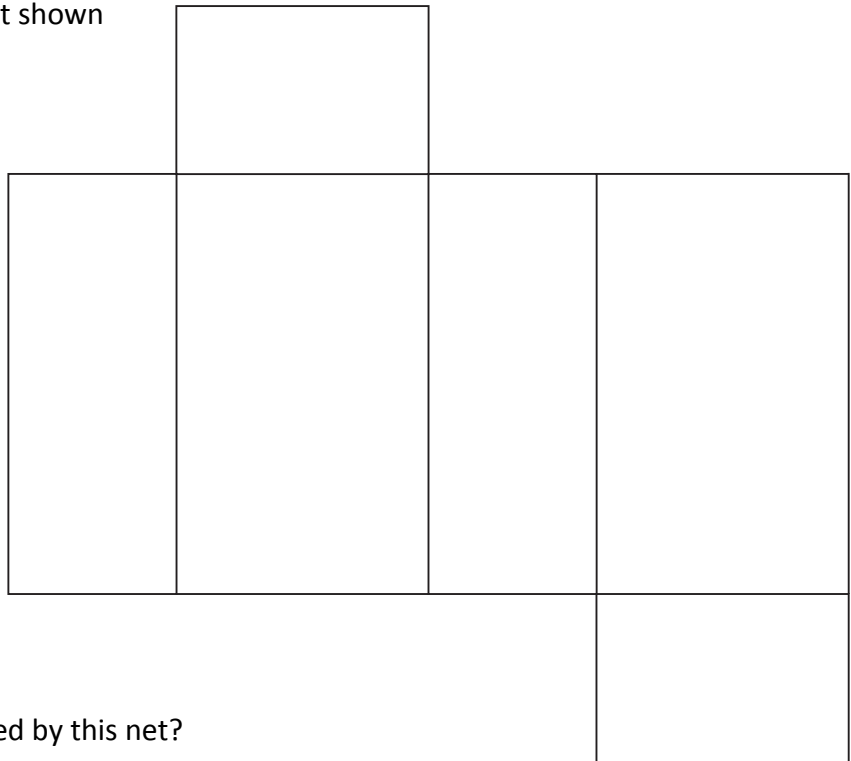


- (d) Find the **area** of the part of the logo that is white.



Question 3 (Suggested maximum time: 5 minutes)

The lengths of the sides in the net shown are 4 cm, 6 cm and 10 cm.



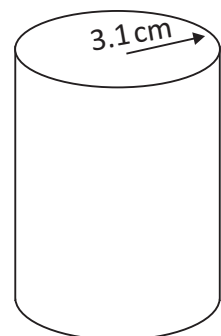
- (a) What 3D solid can be formed by this net? Justify your answer.

[illegible]

- (b)** Find the **volume** of the 3D solid.

[illegible]

- (c) The cylinder shown, with a radius of 3.1 cm, has the same volume as the 3D solid. Find the **height** of the cylinder.
Give your answer correct to the nearest cm.

[illegible]

Question 4 (Suggested maximum time: 10 minutes)

Adam plans to go to the 2018 World Cup football tournament in Russia during the summer.
He has booked his flight, which departs London Heathrow at 21:35 and arrives in Moscow, Russia, 3 hours and 45 minutes later.



- (a)** What time will it be in London when he arrives in Moscow?

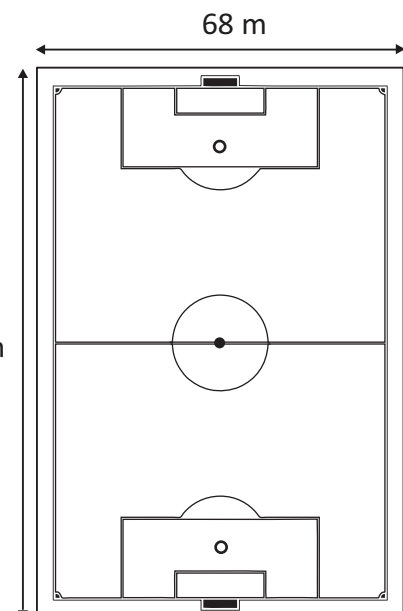
[illegible]

- (b)** Adam is advised to check in at Heathrow airport 150 minutes before his flight departs. What time will he check in at the airport?

[illegible]

The World Cup final will take place on 15th July 2018 at the Luzhniki Stadium in Moscow.

The pitch in this stadium measures 105 m by 68 m.



- (c) Find the length of the **perimeter** of the pitch.

This image shows a full-page sheet of graph paper. It consists of a uniform grid of small squares arranged in 10 columns and 8 rows. The grid covers most of the page, leaving margins at the top, bottom, and sides. There are no markings or text other than the grid itself.

- (d)** Official ticket prices for the group matches and the round of 16 matches at the 2018 World Cup in Russia are listed below.

\$185	\$550	\$105	\$210	\$390	\$220	\$165	\$115	\$245
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- (i) Find the **median** price of the ticket prices listed.

[illegible]

- (ii) Find the **range** of the ticket prices listed.

[illegible]

- (iii) Work out the **mean** price of the different ticket prices.

[illegible]

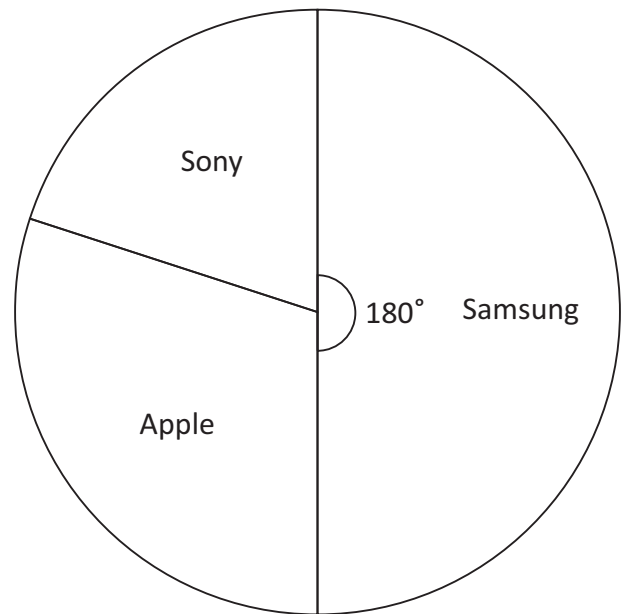
Question 5 (Suggested maximum time: 15 minutes)

Question 5 (Suggested maximum time: 15 minutes)

A shop sells the following brands of smartphones:
Apple, Samsung and Sony.

Last week the shop sold 90 smartphones. The pie chart shows the sales for each of the different brands of smartphone.

There were twice as many *Apple* smartphone sales as *Sony* smartphone sales.



- (a) How many *Samsung* smartphones were sold in the shop last week? Justify your answer fully.

Answer =

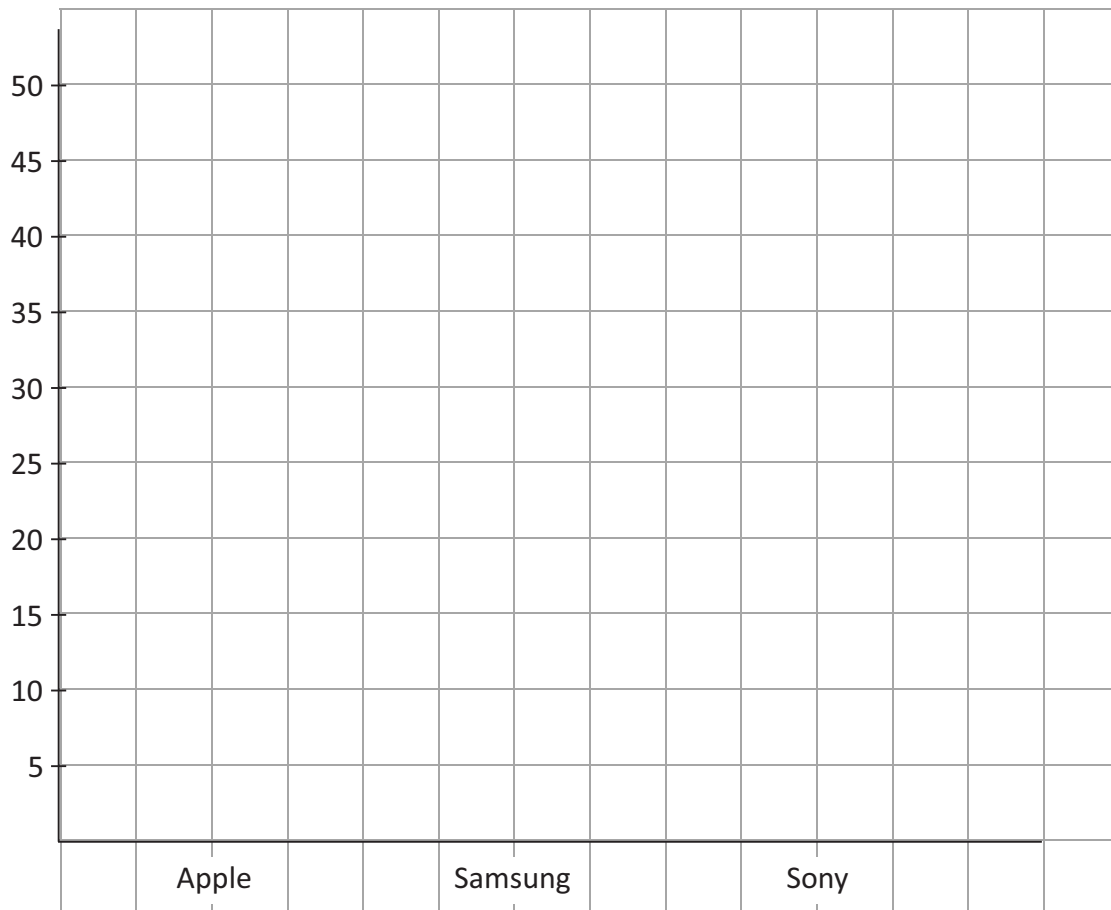
Samsung smartphones

Justification:

[illegible]

- (b)** Cian says “15 *Sony* smartphones were sold in the shop last week”. Show that he is correct.

(c) Draw a bar chart to show the data.



(d) Cian plans to carry out a survey to find out which smartphone people in his local area like best.

(i) Write down two ways in which he can carry out the survey.

[illegible]

(ii) Write a suitable question that Cian could use in his survey.

[illegible]

Question 6 (Suggested maximum time: 10 minutes)

There are black counters, green counters, blue counters and red counters in a bag. Karen is going to pick one counter at random from the bag.

The table below shows each of the probabilities that Karen will pick a black counter or a green counter or a red counter.



Colour	Black	Green	Blue	Red
Probability	0.5	0.25		0.1

- (a)** Find the probability that Karen will pick a **blue** counter.

[illegible]

- (b)** There are 140 counters in the bag.

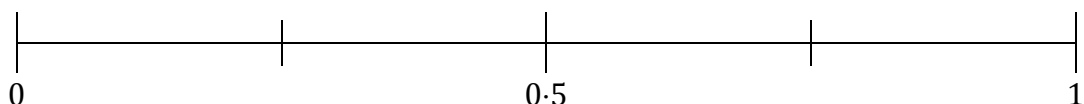
- (i) Find the number of **green** counters in the bag.

[illegible]

- (ii) Find the number of **red** counters in the bag.

[illegible]

- (c) **Mark and label** with the letter **X** the probability that the counter that Karen picked is **not** green on the scale below.

[illegible]

- (d) Another bag contains pink, orange, yellow and white counters. The table below shows the probability of picking each of the different colours at random.
- (i) Fill in the 4 missing values in the table below, to show the probability of picking each colour as a fraction, a percentage and a decimal.
Write the fraction in its simplest form.

Colour	Probability		
	Fraction	Percentage	Decimal
Pink	$\frac{9}{20}$	45%	0.45
Orange	$\frac{3}{10}$		0.3
Yellow	$\frac{1}{5}$	20%	
White		5%	

A large grid of graph paper with 20 columns and 15 rows. The grid is composed of small squares, with a slightly larger margin at the top for writing.

- (ii)** Karen says “there is a 50:50 chance of picking a yellow counter or an orange counter out of the bag”.
- Is she correct?
- Give a reason for your answer.

Answer =

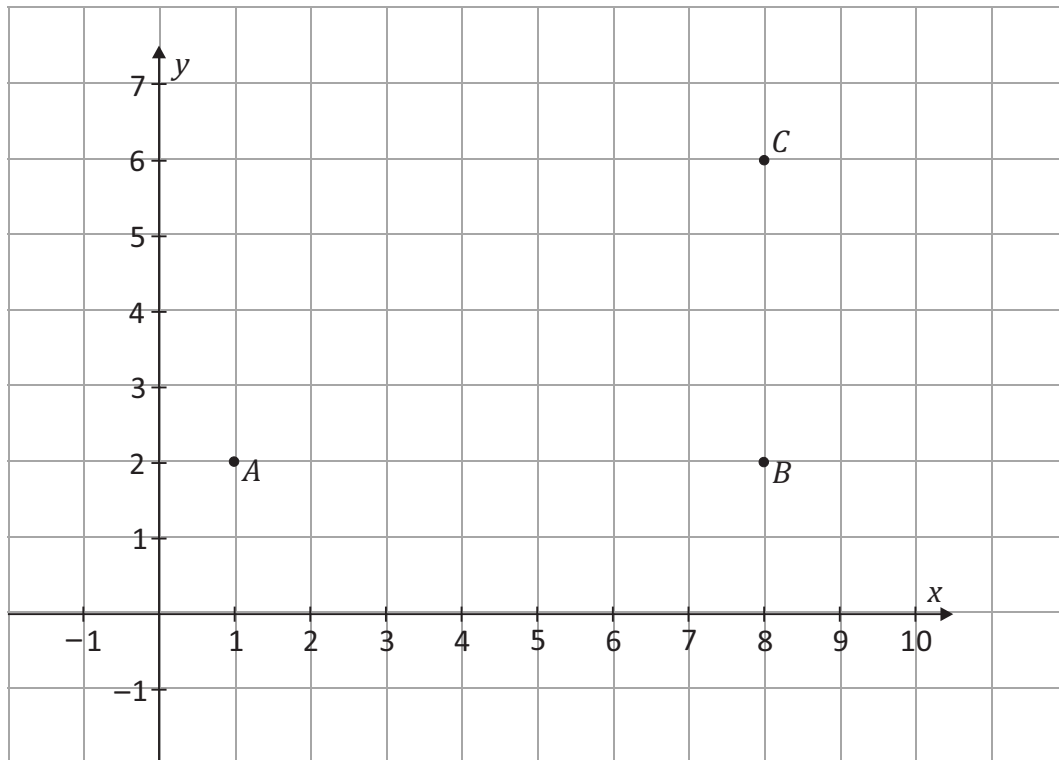
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[illegible]

Question 7

(Suggested maximum time: 15 minutes)

The points A , B , and C are shown on the co-ordinate grid below.



- (a) (i) Write down the co-ordinates of the points A , B and C .

$$A = (\quad , \quad)$$

$$B = (\quad , \quad)$$

$$C = (\quad , \quad)$$

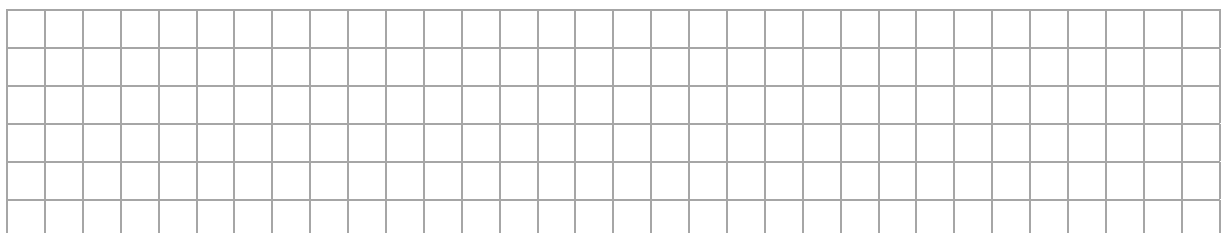
- (ii) $ABCD$ is a rectangle. Plot the point D on the co-ordinate grid and hence write down the co-ordinates of the point D .

$$D = (\quad , \quad)$$

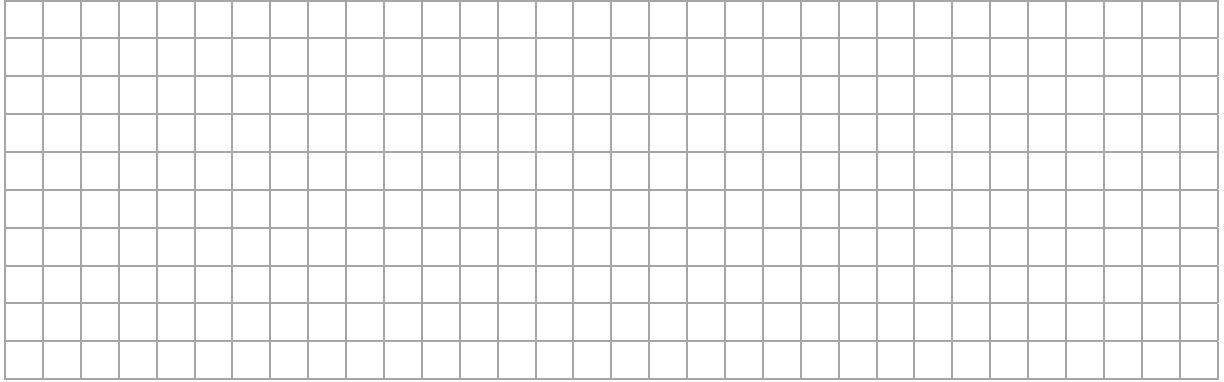
- (iii) On the co-ordinate grid, **draw** the diagonal $[AC]$ of the rectangle $ABCD$.

- (b) Hence, find the co-ordinates of M , the **midpoint** of $[AC]$.

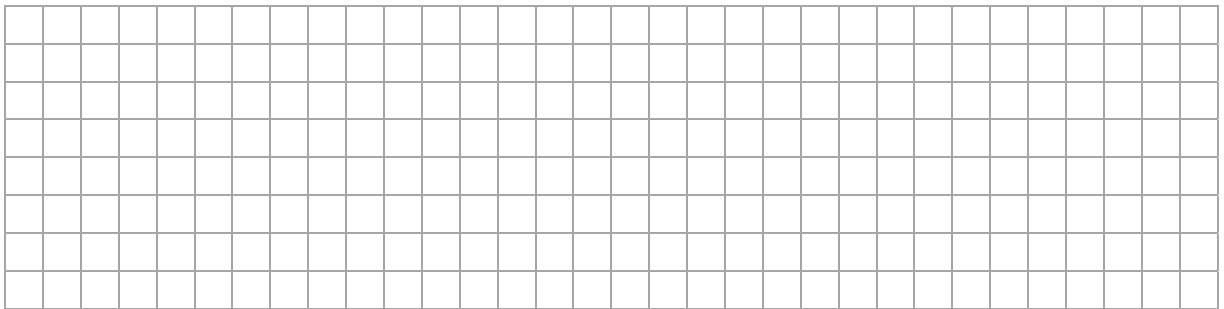
$$M = (\quad , \quad)$$



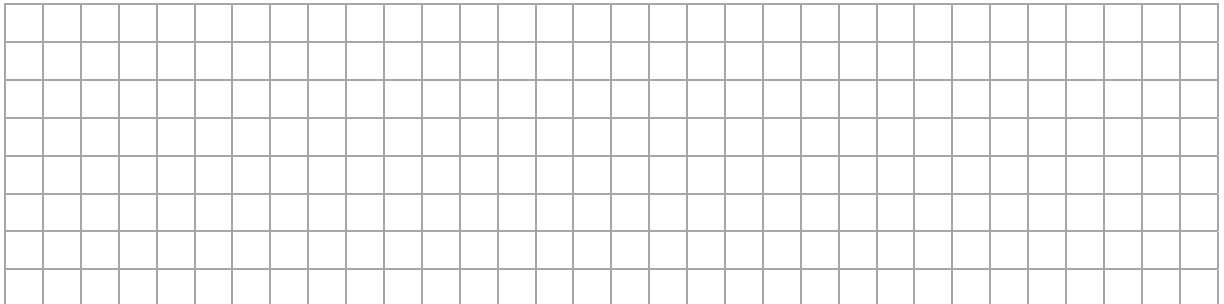
- (c) (i) Find the **length** of $[AC]$. Give your answer in the form $\sqrt{x}$, $x \in \mathbb{N}$.



- (ii) Hence, or otherwise, find the length of the **perimeter** of the triangle ABC .
Give your answer correct to one decimal place.



- (d) Find the **area** of the triangle ABC .



- (e) (i) How many **axes of symmetry** does the rectangle $ABCD$ have?

Answer =

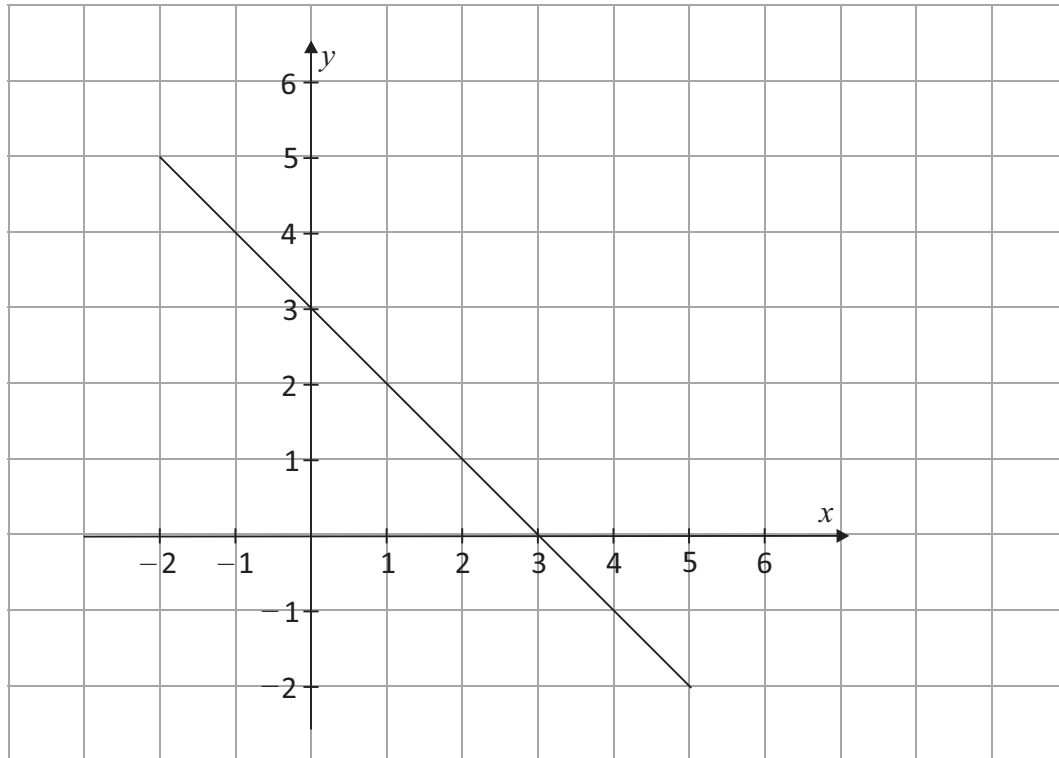
- (ii) Draw the axes of symmetry of the rectangle $ABCD$ on the co-ordinate grid on **page 12**.

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Question 8

(Suggested maximum time: 10 minutes)

The graph of the line $y = -x + 3$ is shown on the co-ordinated grid below.

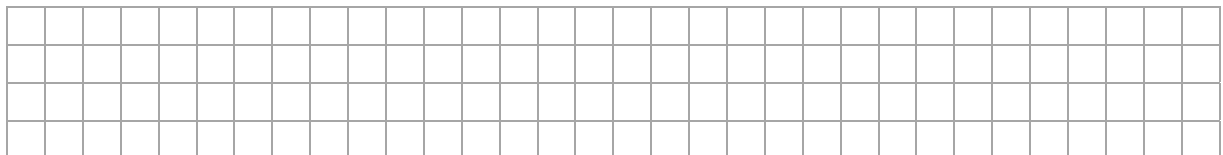


- (a) Write down the co-ordinates of the point where the line crosses the y - axis.

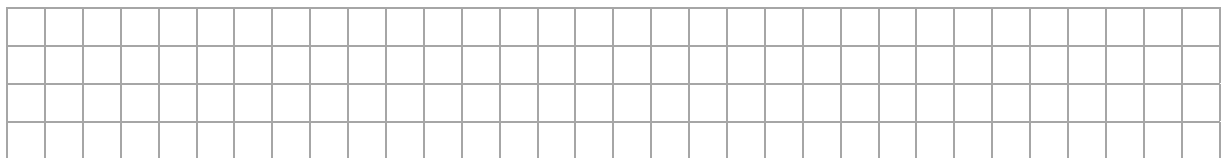
Answer =

- (b) Write down the slope of the line.

Slope =

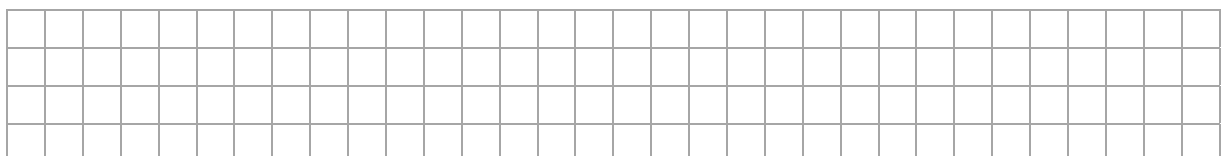


- (c) On the co-ordinate grid, **draw** the graph of the line $y = x - 1$.



- (d) From the graphs or otherwise, find the point of intersection of the two lines.

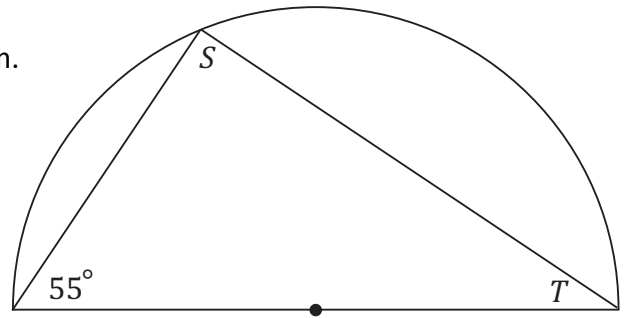
Answer =



Question 9 (Suggested maximum time: 5 minutes)

Question 9 (Suggested maximum time: 5 minutes)

- (a) Theorems on your course can be used to find the measures of the angles in the diagram.



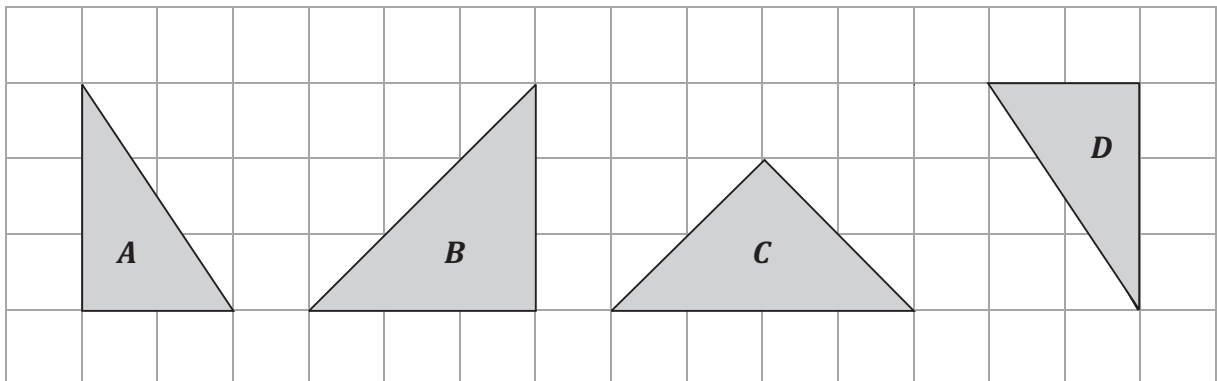
- (i)** Write down, in your own words, any theorem you could use to find $|\angle S|$ or $|\angle T|$.

Theorem:

- (ii)** Find $|\angle S|$ and $|\angle T|$ in the diagram above.

$$|\angle S| =$$
$$|\angle T| =$$

- (b)** Four triangles, A , B , C and D , are shown on the grid below.



Which two triangles are **congruent**?

Answer =

and

Give a reason for your answer.

Reason:

Question 10

(Suggested maximum time: 10 minutes)

The triangle XYZ has:

- one side that is 7 cm long
one angle of 60°
one angle of 45°

- (a)** Work out the size of the **third angle** of the triangle XYZ .

[illegible]

- (b)** What type of triangle is XYZ ? Give a reason for your answer.

Type of triangle:

Equilateral

Isosceles

Scalene

(Tick (✓) one box only)

☐

Give a reason for your answer.

[illegible]

- (c) Construct** the triangle XYZ . Show all of your construction lines clearly.

Construction:

- (d)** Measure the smallest side in your triangle from part **(c)**.
Write the length of this side in the box below, correct to the nearest cm.

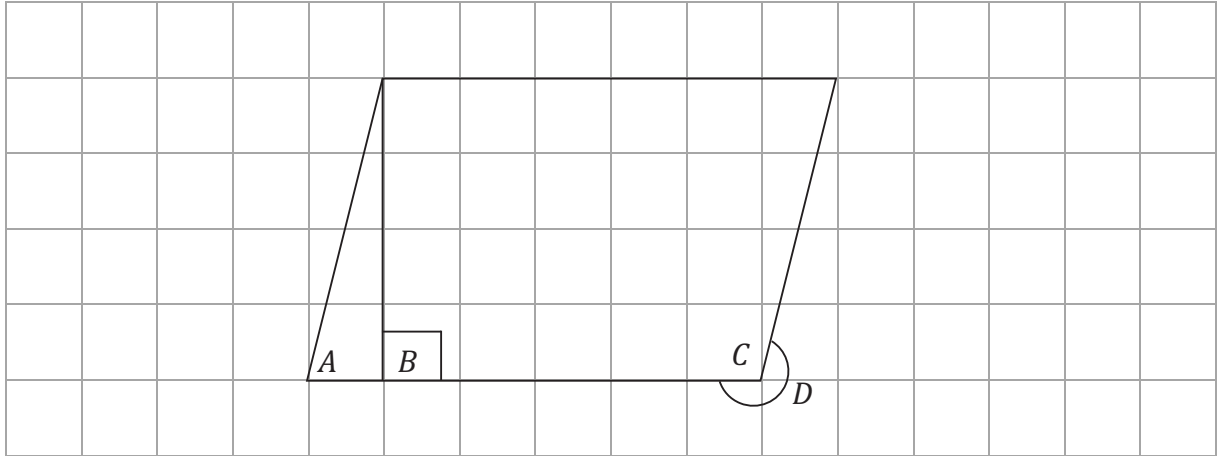
Length of smallest side =

cm

Question 11

(Suggested maximum time: 5 minutes)

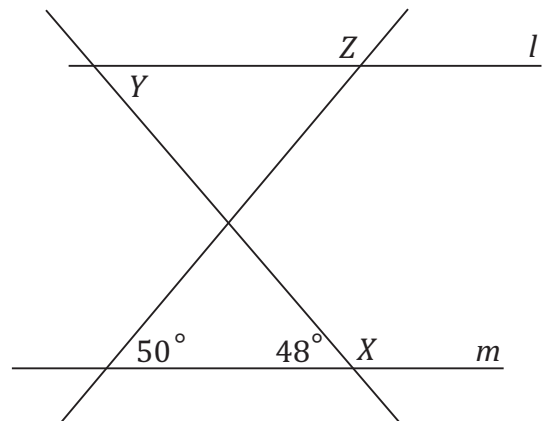
- (a)** Four angles, *A*, *B*, *C* and *D*, are marked on the diagram below.



Using this diagram, write the letter which represents each of the angles listed in the table below.

Type of Angle	Letter
Acute Angle	
Right Angle	
Reflex Angle	
Obtuse Angle	

- (b)** In the diagram, line l is parallel to line m .
The sizes of some of the angles are marked.



Find the size of the angle X , the angle Y and the angle Z , without measuring.

$$|\angle X| = \boxed{} \qquad |\angle Y| = \boxed{} \qquad |\angle Z| = \boxed{}$$

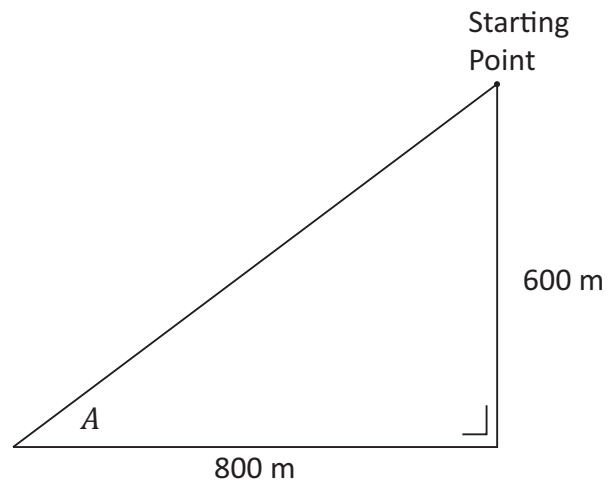
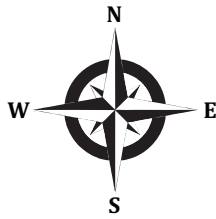
[illegible]

Question 12

(Suggested maximum time: 10 minutes)

Siobhan walks a route that is popular with walkers. She walks 600 m due south and then 800 m due west.

A diagram of her route is shown.



- (a) Write down the length of the side **opposite** to the angle A and the length of the side **adjacent** to the angle A .

Opposite =

	m
--	---

Adjacent =

m

- (b)** Write down the value of $\tan A$ as a fraction.

[illegible]

Answer =

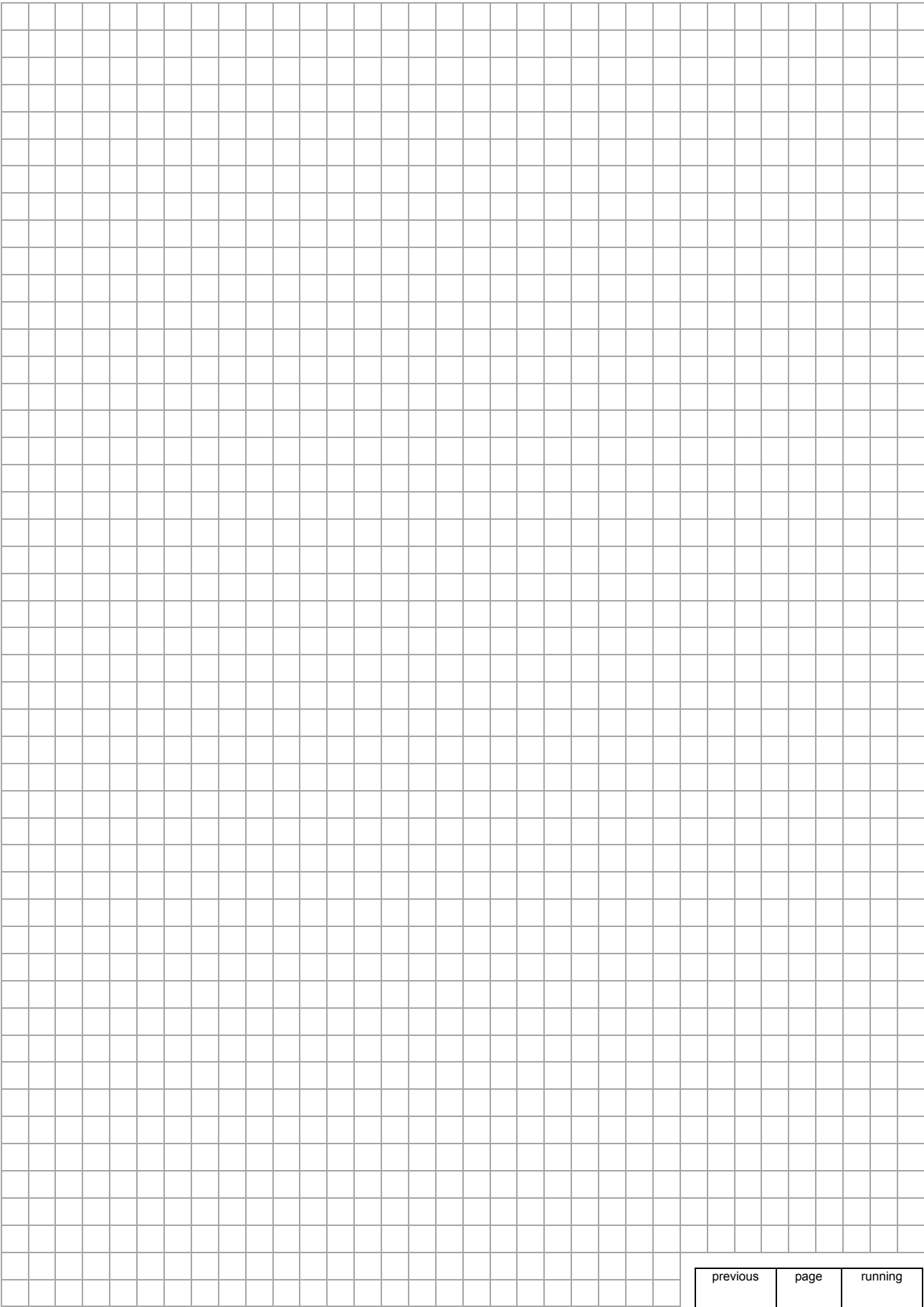
- (c)** Use your answer to part **(b)** to find the **size** of the angle A .
Give your answer correct to the nearest degree.

[illegible]

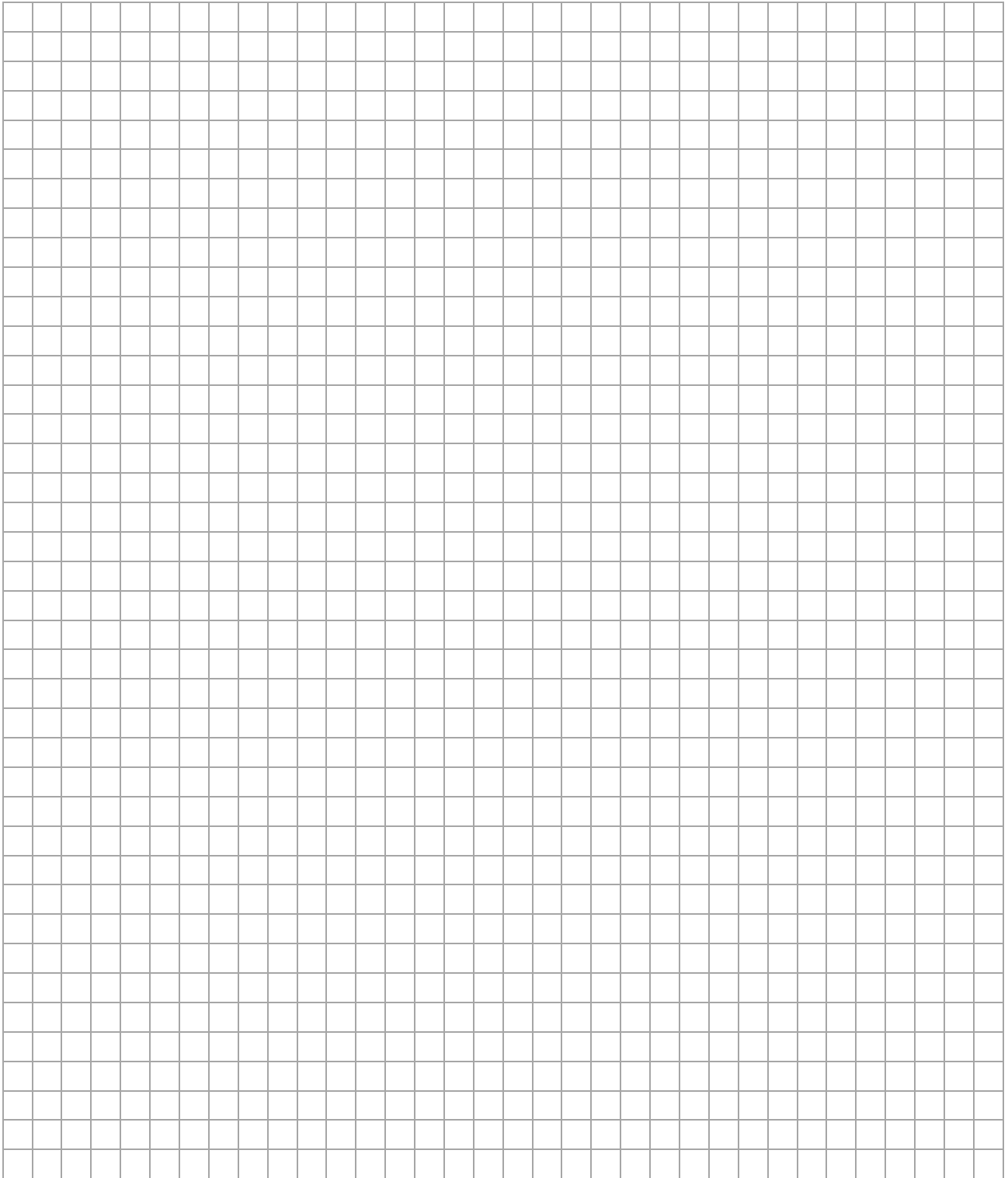
- (d)** Use the theorem of Pythagoras to find the shortest distance Siobhan needs to travel to get back to her starting point. Give your answer in km.
Remember that there are 1000 m in one km.

[illegible]

You may use this page for extra work.



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Pre-Junior Certificate, 2018 – Ordinary Level

Mathematics – Paper 2

Time: 2 hours

